
Machine Learning II

Text Classification

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POLI3148 Data Science in Politics and Public Administration

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Review: What is Machine Learning?

Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data, and thus perform tasks without explicit instructions.

Advances in the field of deep learning have allowed neural networks to surpass many previous approaches in performance.

Source: [Wikipedia](#)

Review: Classification Model

Definition: A model whose prediction is **a class**. For example, the following are all classification models:

- A model that predicts an input sentence's language (French? Spanish? Italian?).
- A model that predicts tree species (Maple? Oak? Baobab?).
- A model that predicts the positive or negative class for a particular medical condition.
- *A model that predicts if an official can get promotion*
- *A model that predicts if a war will happen*
- *A model that predicts which candidate can win the election*

In contrast, **regression models** predict numbers rather than classes.

Source: [Google Machine Learning](#)

Text Classification Model

Definition: A ML model whose **input (X)** is text and predicted **target (y)** is a **class**.
For example, the following are all classification models:

A model predicting...

- if a social media post contains hate speech
- if a official document discusses political violence
- if a question in a preference conference contains positive/ negative/ neutral sentiment
- if a question to ChatGPT is inappropriate, for which ChatGPT should refuse to answer

Workflow

Data Input:

- Load data
- Define **X** and **y**
- Handle missing values
- Split data into training and test sets
- Re-formats for Machine Learning tool

Data Processing:

- Define model
- Train model

Data Output:

- Evaluate model
- Interpret model
- Discussion: What your results say about the substantive topics



**How Text Classification is
special:**

**Define X and Y
Interpret model**

Running Example: Understanding Chinese Diplomacy w/ Text Analysis

Chinese Ministry of Foreign Affairs Press Conferences Corpus

- Data: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/BAKGET>
- Paper: <https://link.springer.com/article/10.1007/s11067-020-09583-8>

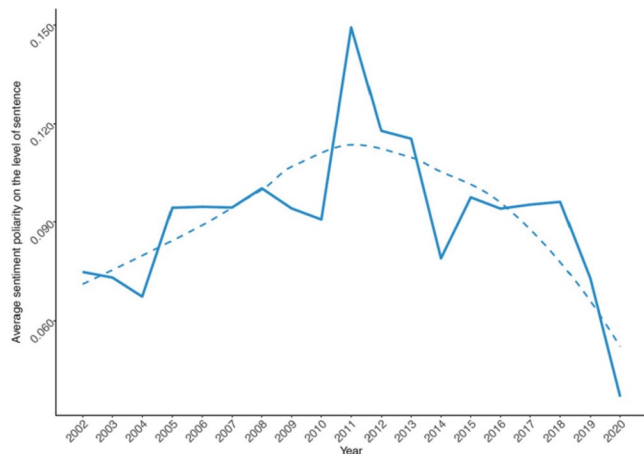


Fig. 4 Change in sentiment polarity towards the US in Chinese foreign policy discourse. Note: Solid line visualizes raw sentiment averages on the level of sentences. Dashed line visualizes smoothed trend (using loess function [Local Polynomial Regression Fitting])



I AM BACK!

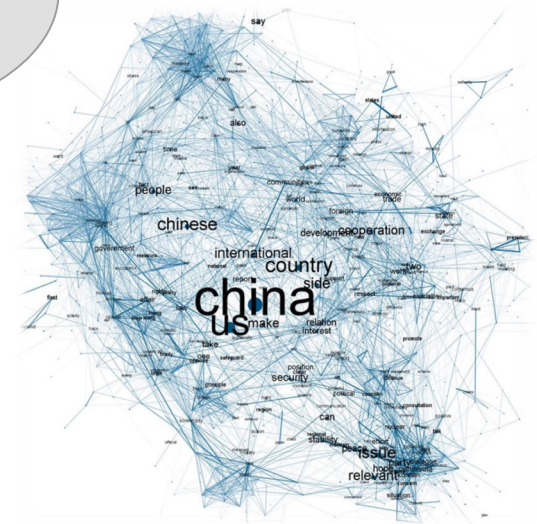


Fig. 6 Semantic network tracing US-China discourse under President Xi Jinping

What Have We Done with These Data?

- Extracting Textual Features
 - Linguistic features
 - Tokenization
 - Lemmatization
 - Stopword removal
 - Named Entities
- Topic Modeling

We used a bag-of-word approach to extract textual features and conducted some macro-level (corpus-level) analysis

We have not done any **micro-level** analysis



Text Classification Application: Sentiment Analysis

Task:

Develop a ML classifier to identify “aggressive” questions in the MoFA Q&A dataset

Steps:

- Manually label a small portion of the MoFA Q&A
 - Train and evaluate a ML classifier
 - Apply the ML classifier to Q&A that are not manually labeled
- 

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Get y

Get **y** from the raw data

- Create a “**q_aggressive**” column using the existing “**q_sentiment**” column
 - Normally, y is obtained through manual labeling
 - We take advantage of the dataset, which obtain the sentiment of questions using other machine learning methods
 - For this task, we define it as questions whose sentiment scores are at the bottom 25 percentile

Get X

Get **X** from the linguistic feature: Load the linguistic features we previously extracted from questions using *spacy*

- Remove “useless” tokens (stopwords, punctuations, numbers, etc.)
- Reshape the table to make one row per document (a *document* is a question in our case)
- Create a **document-term matrix** using CountVectorizer

Remember this?
See Lecture 8



	she	loves	pizza	is	delicious	a	good	person	people	are	the	best
She loves pizza, pizza is delicious	1	1	2	1	1	0	0	0	0	0	0	0
She is a good person	1	0	0	1	0	1	1	1	0	0	0	0
good people are the best	0	0	0	0	0	0	1	0	1	1	1	1

Follow the code.

**See how to obtain y , X ,
 X_names**

The rest are almost the same

Workflow

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Data Processing:

- Define model
- Train model

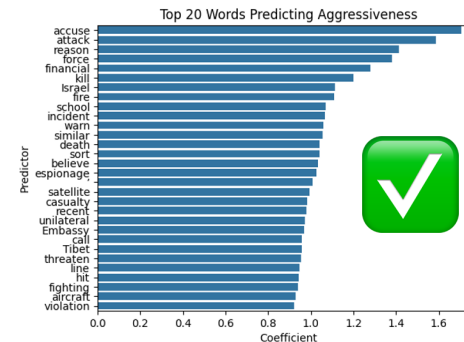
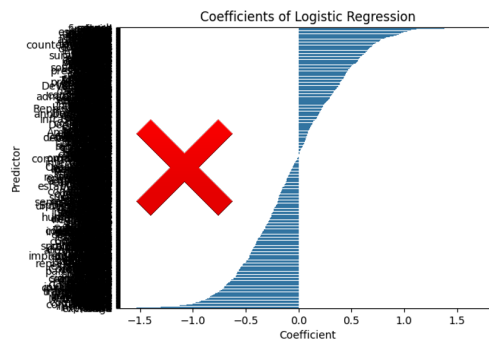
Data Output:

- Evaluate model
- **Interpret model**
- Discussion: What your results say about the substantive topics



Interpret Model: Be Selective

- With text classification, or other classification that contains a lot of predictors, you cannot interpret ALL predictors (thousands of them)
- What to do
 - Interpret only a subsample of predictors
 - Sort the coefficients/ feature importance
 - For logistic regression: Interpret those with largest coefficients
 - For Decision Trees/ Random Forest: Interpret those with highest feature importance



**Try Decision Trees and
Random Forest yourself**

Finally,

A Postmodern Approach to Machine Learning

Remember Hugging Face?

Source: https://huggingface.co/models?pipeline_tag=text-classification&sort=trending

The screenshot shows the Hugging Face website interface. At the top, there's a navigation bar with the Hugging Face logo, a search bar, and links to Models, Datasets, Spaces, Posts, Docs, Enterprise, and Pricing. Below the navigation bar, a yellow banner reads "Hugging Face is way more fun with friends and colleagues! 🥳 Join an organization".

The main content area is divided into two columns. The left column contains a sidebar with navigation tabs: Tasks, Libraries, Datasets, Languages, Licenses, and Other. The "Tasks" tab is selected, showing a search bar and a "Reset Tasks" button. Below this, there are two sections: "Multimodal" and "Computer Vision". The "Multimodal" section lists tasks like Audio-Text-to-Text, Image-Text-to-Text, Visual Question Answering, Document Question Answering, and Video-Text-to-Text. The "Computer Vision" section lists tasks like Depth Estimation, Image Classification, Object Detection, Image Segmentation, Text-to-Image, Image-to-Text, Image-to-Image, Image-to-Video, Unconditional Image Generation, Video Classification, Text-to-Video, Zero-Shot Image Classification, Mask Generation, Zero-Shot Object Detection, Text-to-3D, Image-to-3D, Image Feature Extraction, and Keypoint Detection. Below these is the "Natural Language Processing" section, which lists tasks like Text Classification, Token Classification, Table Question Answering, and Question Answering.

The right column displays a list of models. At the top, there's a "Models" header with a count of 73,502 and a "Filter by name" search bar. Below this, there are two buttons: "Full-text search" and "Sort: Trending". The list of models is displayed in a grid format, showing the model name, its pipeline tag, and its update date. The models listed are:

- cardiffnlp/twitter-roberta-base-sentiment-latest**: Text Classification, Updated May 28, 2023, 4.89M, 579 likes.
- distilbert/distilbert-base-uncased-finetuned-sst-2-...**: Text Classification, Updated Dec 20, 2023, 9.43M, 626 likes.
- SamLowe/roberta-base-go_emotions**: Text Classification, Updated Oct 4, 2023, 1.65M, 481 likes.
- BAAI/bge-reranker-v2-m3**: Text Classification, Updated Jun 24, 680k, 394 likes.
- hbseong/HarmAug-Guard**: Text Classification, Updated Oct 14, 677, 32 likes.
- ProsusAI/finbert**: Text Classification, Updated May 23, 2023, 1.75M, 683 likes.
- HuggingFaceFW/fineweb-edu-classifier**: Text Classification, Updated 8 days ago, 23.7k, 136 likes.
- Qwen/Qwen2.5-Math-RM-72B**: Text Classification, Updated 25 days ago, 9.46k, 55 likes.
- j-hartmann/emotion-english-distilroberta-base**: Text Classification, Updated Jan 2, 2023, 987k, 355 likes.
- PleIAs/celadon**: Text Classification, Updated 22 days ago, 278, 16 likes.
- allenai/Llama-3.1-Tulu-3-8B-RM**: Text Classification, Updated 4 days ago, 271, 5 likes.
- microsoft/MiniLM-L12-H384-uncased**: Text Classification, Updated May 20, 2021, 15.2k, 83 likes.
- papluca/xlm-roberta-base-language-detection**: Text Classification, Updated Dec 28, 2023, 3.56M, 283 likes.
- ElKulako/cryptobert**: Text Classification, Updated Jan 31, 9.8k, 104 likes.
- mzm8488/distilroberta-finetuned-financial-news-sent...**: Text Classification, Updated Jan 21, 1.49M, 343 likes.
- pysentimiento/robertuito-sentiment-analysis**: Text Classification, Updated Jul 9, 1.03M, 78 likes.

Using Resources from Hugging Face

- A playground of ML models... including text classification models
- Pros
 - State-of-the-art: Latest published works in ML and AI publish their models here
 - User-friendly
- Cons
 - Not all are reliable, need to be selective (if unsure, use the most popular ones)
 - Some are black boxes (in terms of model architecture and training data)
 - Authors can delete the resources (don't assume it will always be available.)

**Let's try a simple text
classification task together**